6.1.6 Wrap-Up: Cool Down

1. Explain how dilations are different from rigid transformations, in your own words.

Unit 6, Lesson 2: Sequences of Transformations That Include Dilations

Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

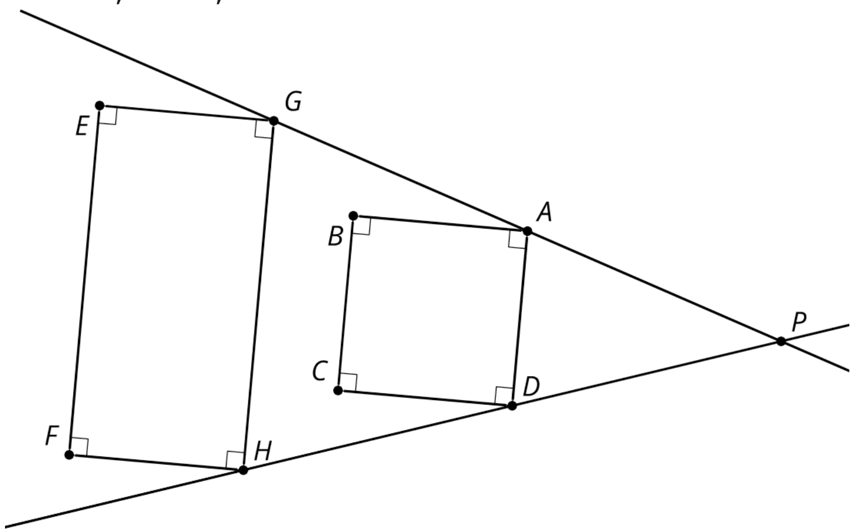
MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

Learning Targets

- ☐ I can describe a sequence of transformations that includes a dilation.
- ☐ I can represent a sequence of transformations using coordinate notation with algebraic descriptors.

6.2.1 Warm-Up: Dilation Miscalculation

1. Cassandra tried to determine if the diagram shown represents a dilation.
   $EGHF$, $BADC$, $\overleftrightarrow{PG}$ and $\overleftrightarrow{PH}$ are shown.



Cassandra listed four characteristics of dilations:

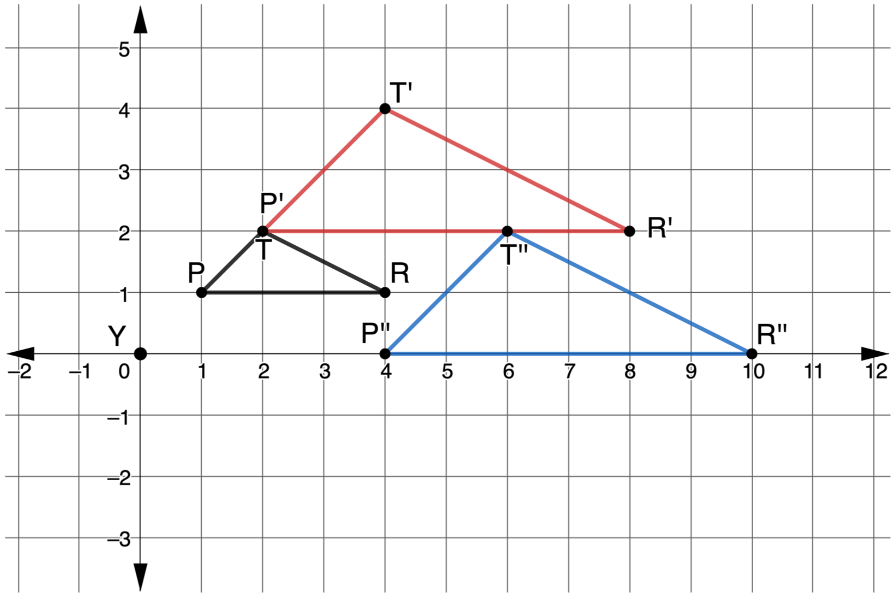
Characteristic	Yes/No
Corresponding angle measures are equal	Yes
Corresponding sides are parallel	Yes
Corresponding vertices are collinear with the point of dilation	Yes
Orientation of the shapes are the same	Yes

a. Of the characteristics Cassandra listed, which are not true for $EGHF$ and $BADC$?

b. Explain why the diagram shows that the dilation does not have these characteristics. If necessary, use the diagram to draw.

6.2.2 Guided Instruction: Sequence of Transformations

1. A sequence of transformations of ΔPTR is shown on the coordinate grid. ΔPTR is dilated then translated, where $\Delta P'T'R'$ is the dilation and $\Delta P''T''R''$ is the translation of $\Delta P'T'R'$.



Triangle PTR	
P	$(1, 1)$
T	$(2, 2)$
R	$(4, 1)$

Triangle $P'T'R'$	
P'	$(2, 2)$
T'	$(4, 4)$
R'	$(8, 2)$

Triangle $P''T''R''$	
P''	$(4, 0)$
T''	$(6, 2)$
R''	$(10,0)$

a. Complete the description.

ΔPTR was dilated using a scale factor of ______ centered at the origin. The resulting polygon, $\Delta P'T'R'$, was translated ________________________________ to create $\Delta P''T''R''$.

b. Complete the table.

Transformation	Algebraic Description
Dilation	$(x, y) \rightarrow$ (_____x, _____y)
Translation	$(x, y) \rightarrow$ (x______, y______)

c. Determine the coordinates of each triangle if ΔPTR is first translated then dilated.

Pre-image

Triangle PTR	
P	$(1, 1)$
T	$(2, 2)$
R	$(4, 1)$

Translation

Triangle $P'T'R'$	
P'	
T'	
R'	

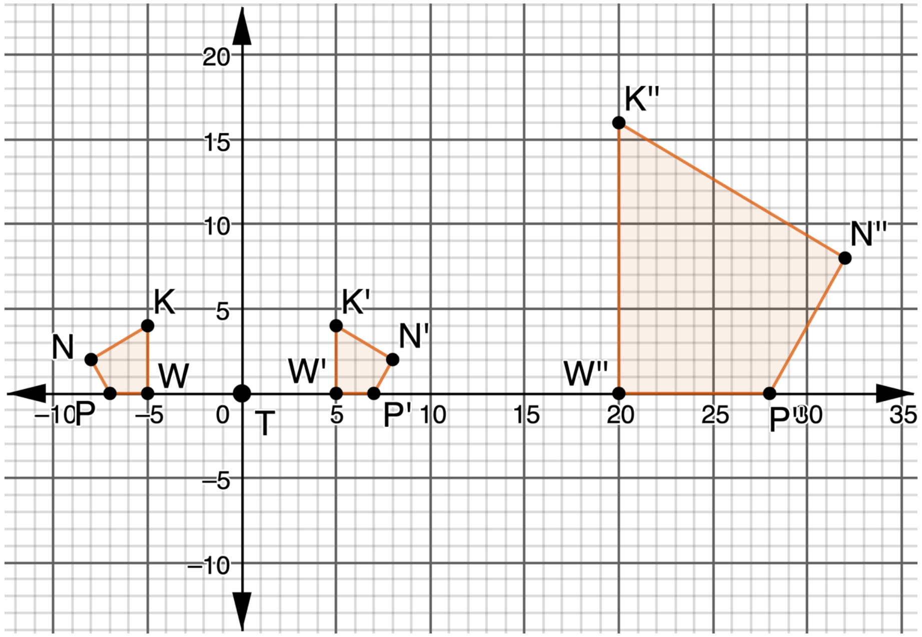
Dilation

Triangle $P''T''R''$	
P''	
T''	
R''	

Some sequences of transformations are commutative, which means the transformations can be applied in any order.

d. Use the tables to determine if the sequence of transformations applied to ΔPTR is commutative. Work with your partner to write an explanation.

2. $KWPN$, $K'W'P'N'$, and $K''W''P''N''$ are shown on the coordinate grid.



$KWPN$	
K	$(-5, 4)$
W	$(-5, 0)$
P	$(-7, 0)$
N	$(-8, 2)$

$K'W'P'N'$	
K'	$(5, 4)$
W'	$(5, 0)$
P'	$(7, 0)$
N'	$(8, 2)$

$K''W''P''N''$	
K	$(20, 16)$
W	$(20, 0)$
P	$(28, 0)$
N	$(32, 8)$

a. Discuss with your partner the first transformation that occurred. Describe how you determined the first transformation.

b. Determine the scale factor in the second transformation.

c. Complete the description of the transformations.

$KWPN$ was ________________________________ to create $K'W'P'N'$. $K'W'P'N'$ was dilated using a scale factor of ______ centered at _______ to create $K''W''P''N''$.

d. Complete the table.

Transformation	Algebraic Description
	$(x, y) \rightarrow$
	$(x, y) \rightarrow$

e. Work with your partner to determine if the sequence of transformations applied to $KWPN$ is commutative. Complete the table by first applying the dilation, and then the other transformation. Then, write a conclusion based on the tables.

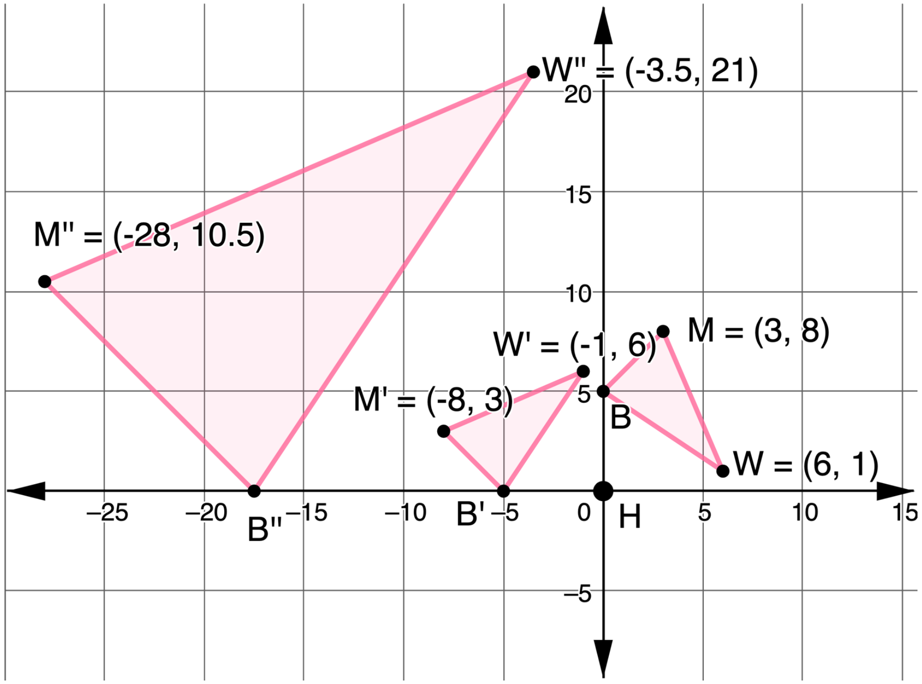
$KWPN$	
K	$(-5, 4)$
W	$(-5, 0)$
P	$(-7, 0)$
N	$(-8, 2)$

$K'W'P'N'$	
K'	
W'	
P'	
N'	

$K''W''P''N''$	
K	
W	
P	
N	

6.2.3 Guided Practice: Mapping a Dilation

1. A sequence of transformations of ΔMWB is shown on the coordinate grid.



a. Complete the table of values for each triangle.

Preimage **Dilation**

ΔMWB	
M	$(3, 8)$
W	$(6, 1)$
B	

$\Delta M'W'B'$	
M'	$(-8, 3)$
W'	$(-1, 6)$
B'	

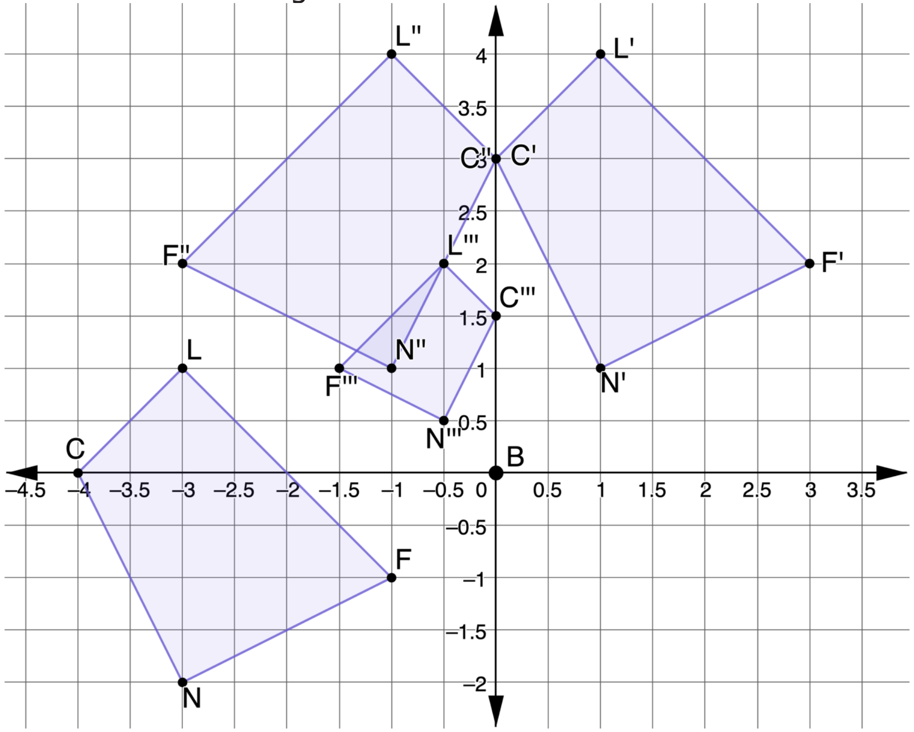
$\Delta M''W''B''$	
M''	$(-28, 10.5)$
W''	$(-3.5, 21)$
B''	

b. Complete the table.

Transformation	Algebraic Description
	$(x, y) \to$
	$(x, y) \to$

6.2.4 Your Turn!

1. $LFNC$, $L'F'N'C'$, $L''F''N''C''$, and $L'''F'''N'''C'''$ are shown on the coordinate grid.



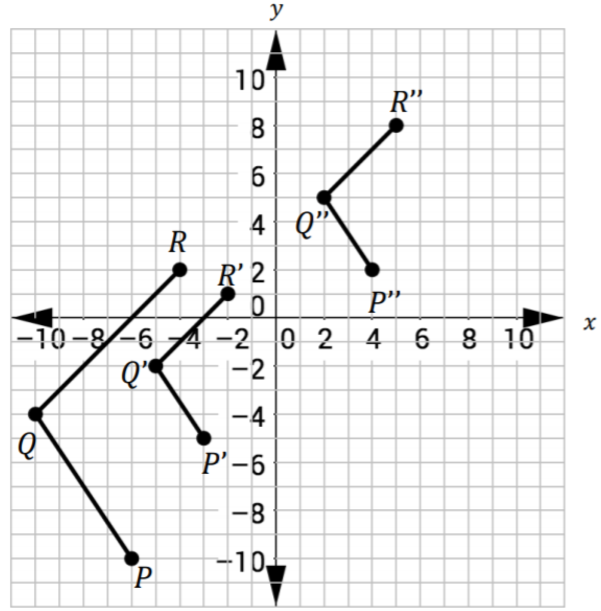
a. Describe the sequence of transformations of $LFNC$ that resulted in $L'''F'''N'''C'''$.

b. Write the algebraic descriptions of the sequence of transformations.

Transformation	Algebraic Description
	$(x, y) \rightarrow$
	$(x, y) \rightarrow$
	$(x, y) \rightarrow$

6.2.5 Wrap-Up: Ticket Out the Door

1. Points $P(-6, -10)$, $Q(-10, -4)$, and $R(-4, 2)$ are plotted on the coordinate grid to form figure PQR. Two transformations of figure PQR are shown.



Gladys first transformed figure PQR to $P'Q'R'$ using a dilation centered at the origin. She then transformed figure $P'Q'R'$ to $P''Q''R''$.
Complete the table.

Transformation	Written Description	Algebraic Description
PQR to $P'Q'R'$		$(x, y) \rightarrow$
$P'Q'R'$ to $P''Q''R''$		$(x, y) \rightarrow$